Done

Reverse Index (with stop words) in SQL

In this assignment, you will create a table of documents and then produce a reverse index with stop words for those documents that identifies each document which contains a particular word using SQL.

Connection details:

Host: pg.pg4e.com

Port: 5432

Database: pg4e_402ae32925

User: pg4e_402ae32925

Password: (hide/show copy)

Check Answer

Here is the the **psql** command to access your database:

```
psql -h pg.pg4e.com -p 5432 -U pg4e_402ae32925 pg4e_402ae32925
```

```
CREATE TABLE docs02 (id SERIAL, doc TEXT, PRIMARY KEY(id));

CREATE TABLE invert02 (
  keyword TEXT,
  doc_id INTEGER REFERENCES docs02(id) ON DELETE CASCADE
);
```

If you already have the above tables created and the documents inserted from a prior assignment, you can just delete all the rows from the reverse index and recreate them following the rules of stop words:

```
DELETE FROM invert02;
```

Here are the one-line documents that you are to insert into **docs02**:

```
INSERT INTO docs02 (doc) VALUES
('should be wondering a bit about the Python interpreter itself What'),
('language is it written in Is it written in a compiled language When we'),
('type python what exactly is happening'),
('The Python interpreter is written in a highlevel language called C'),
('You can look at the actual source code for the Python interpreter by'),
('source code So Python is a program itself and it is compiled into'),
('machine code When you installed Python on your computer or the vendor'),
('installed it you copied a machinecode copy of the translated Python'),
('program onto your system In Windows the executable machine code for'),
('Python itself is likely in a file with a name like');
```

Done

Here are your stop words (https://www.ranks.nl/stopwords):

```
CREATE TABLE stop_words (word TEXT unique);

INSERT INTO stop_words (word) VALUES
('i'), ('a'), ('about'), ('an'), ('are'), ('as'), ('at'), ('be'),
('by'), ('com'), ('for'), ('from'), ('how'), ('in'), ('is'), ('it'), ('of'),
('on'), ('or'), ('that'), ('the'), ('this'), ('to'), ('was'), ('what'),
('when'), ('where'), ('who'), ('will'), ('with');
```

Here is a sample for the first few expected rows of your reverse index:

```
SELECT keyword, doc_id FROM invert02 ORDER BY keyword, doc_id LIMIT 10;

keyword    |  doc_id
-----------+--------
actual     |    5
and        |    6
bit        |    1
c          |    4
called     |    4
can        |    5
code       |    5
code       |    6
code       |    7
code       |    9
```